

Alessandro Ristori

MACHINE LEARNING ENGINEER · DATA SCIENTIST

Scandicci (FI), Tuscany, Italy

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About me

I have a Master's degree in Computer Science from the University of Pisa, specializing in Artificial Intelligence. I am interested in exploring the ever-expanding world of Machine Learning, mainly Natural Language Processing and Machine Translation, but not averse to testing my skills in other domains.

Work Experience

TAI Software Solution Srl

BIG DATA & AI SPECIALIST

Pisa (Mainly remote), Italy

May. 2024 - present

- Developed Computer Vision and NLP solutions integrated with IoT and cloud infrastructures.
- Delivered internal training on AI/ML methodologies and tooling.
- Built scalable Big Data pipelines and AI solutions on AWS using managed cloud services.
- Developed AI-driven solutions for business applications using advanced analysis of large-scale data.

Education

University of Pisa

M.Sc. IN COMPUTER SCIENCE - AI CURRICULUM

Pisa, Italy

Sep. 2020 - Dec. 2023

- **Final mark:** 110/110 cum Laude.
- **GPA:** 29.53/30 IT $\approx$ 3.94/4 US.
- **Thesis:** *Continual Learning for Non-Autoregressive Neural Machine Translation.*

University of Florence

B.Sc. IN COMPUTER SCIENCE

Florence, Italy

Sep. 2016 - Feb. 2020

- **Final mark:** 110/110.
- **GPA:** 28.23/30 IT $\approx$ 3.76/4 US.
- **Thesis:** *Genetic Algorithms and their Applications.*

ISISTL Russell-Newton

HIGH SCHOOL

Scandicci, Italy

Sep. 2011 - Jul. 2016

- **Final mark:** 100/100 cum Laude.

Personal Projects

Centered Kernel Alignment in PyTorch

 Repository

SIMILARITY INDEX FOR NEURAL NETWORKS

- Implemented Centered Kernel Alignment in PyTorch. Centered Kernel Alignment (CKA) is a similarity index between representations of features in neural networks, based on the Hilbert-Schmidt Independence Criterion (HSIC).
- **Technical skills:** Python, PyTorch, HuggingFace (transformers), Git.

Continual Learning for Non-Autoregressive Neural Machine Translation

 Repository

M.Sc. THESIS PROJECT

- M.Sc. thesis on tackling the issue of Catastrophic Forgetting on two different non-autoregressive Neural Machine Translation models (CMLM and GLAT) by using the continual strategy of Experience Replay.
- **Technical skills:** Python, PyTorch, HuggingFace (transformers, datasets, tokenizers), PyTorch Lightning, WandB, Git.

Skills

Main Interests	Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), Neural Machine Translation (NMT).
Programming	Python (main), Java, MATLAB, SQL.
AI / ML	PyTorch, HuggingFace (transformers, datasets, tokenizers), PyTorch Lightning, Keras.
Data Science	Pandas, scikit-learn, NumPy, Matplotlib, Apache Spark.
Cloud	Amazon Web Services (AWS).
Tools	Linux, Git, Docker.

Certificates

2026	Associate Data Pipeline Developer Certification , Boomi	 Badge
2026	Professional Integration Developer Certification , Boomi	 Badge
2026	Associate Integration Developer Certification , Boomi	 Badge
2026	AWS Certified Machine Learning Engineer – Associate , Amazon Web Services (AWS)	 Badge
2025	AWS Certified AI Practitioner , Amazon Web Services (AWS)	 Badge

Languages

Italian	Native proficiency
English	Professional working proficiency (C1)
Spanish	Limited working proficiency (B1)

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